

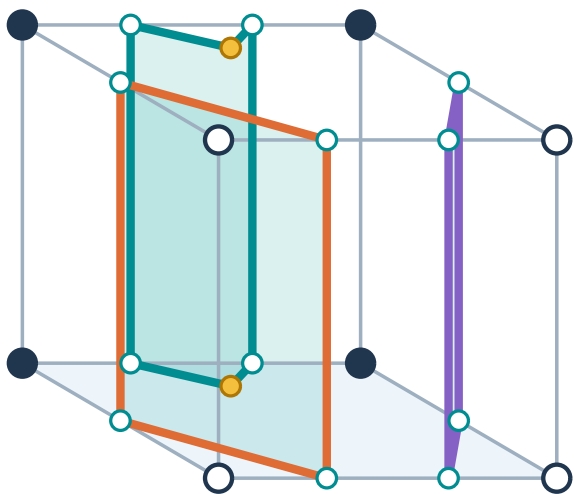
双交点连接图册

23 个代表 case + 7 组内外翻转对照

按项目源码查表规则绘制，所有图形均为可缩放矢量。

CASE 3d

3 loops



face 0x53 counts [2, 1, 2, 1]

cube 0x00F33

如何读图

实心角点表示内部，空心角点表示外部；小圆点是边交点；黄色点是面内转折点。彩色线区分闭合边界环，半透明面表示示意曲面。

每个二维面复制到 $z=0$ 与 $z=1$ ，四条 z 方向边均无交点，六面查表后形成完整立方体。这是面模板的一种三维延伸。

23 是按旋转/镜像归并后的面模板数。展开后有 82 个合法面状态，完整立方体有 36,450 个合法编码。

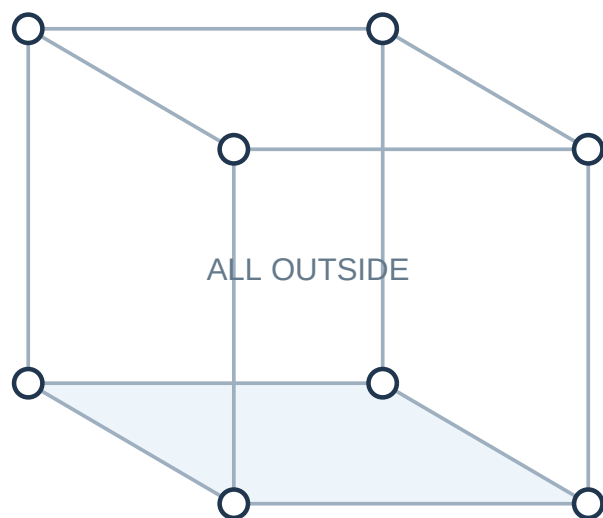
坐标约定：单交点 $t=0.5$ ，双交点 $t=0.32/0.68$ ，黄色点向面内偏移 0.20。几何为示意，不是 GT mesh 重建。

github.com/crazyMessi/MeshUtil

代表模板 / 1a + 1b

CASE 1a

0 loops

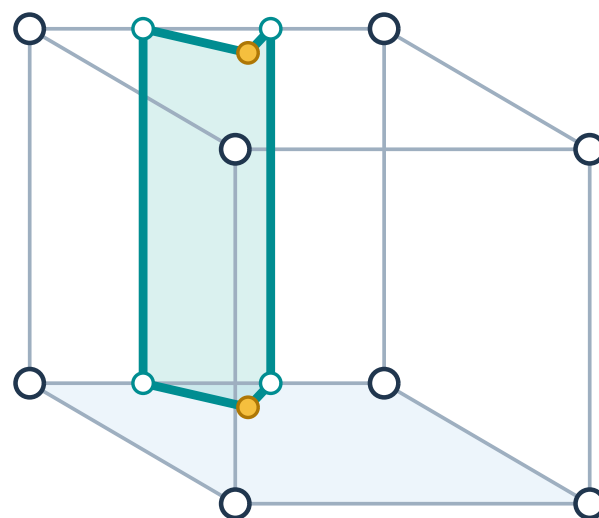


face 0x00 counts [0, 0, 0, 0]

cube 0x00000

CASE 1b

1 loops



face 0x10 counts [2, 0, 0, 0]

cube 0x00500

面角点掩码 0000 | 双交点掩码 0000

闭环节点数: 0

全部外部; 没有边界环。

面角点掩码 0000 | 双交点掩码 0001

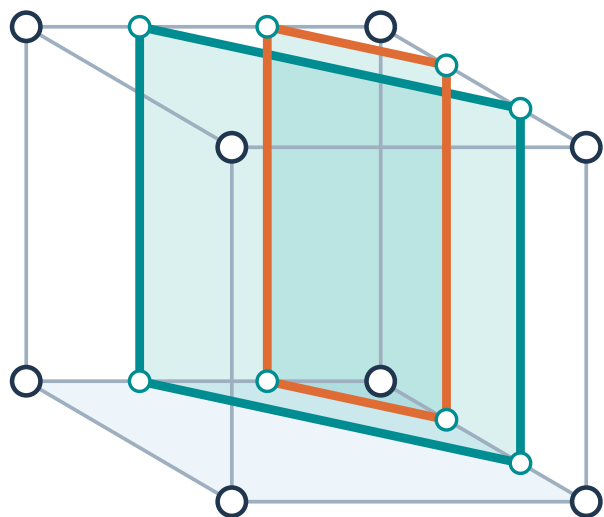
闭环节点数: 6

黄色点为示意转折位置。

代表模板 / $1c + 1d$

CASE 1c

2 loops

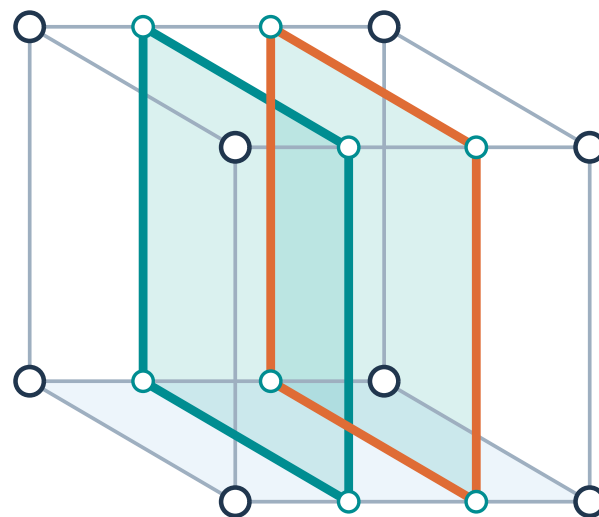


face 0x30 counts [2, 2, 0, 0]

cube 0x0A500

CASE 1d

2 loops



face 0x50 counts [2, 0, 2, 0]

cube 0x00F00

面角点掩码 0000 | 双交点掩码 0011

闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

面角点掩码 0000 | 双交点掩码 0101

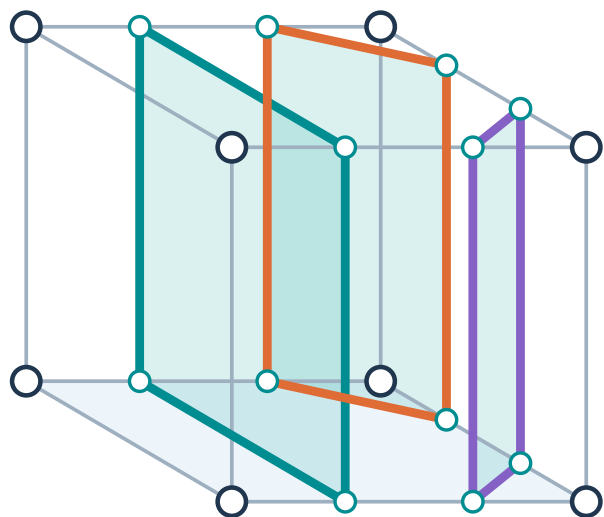
闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / $1e + 1f$

CASE 1e

3 loops

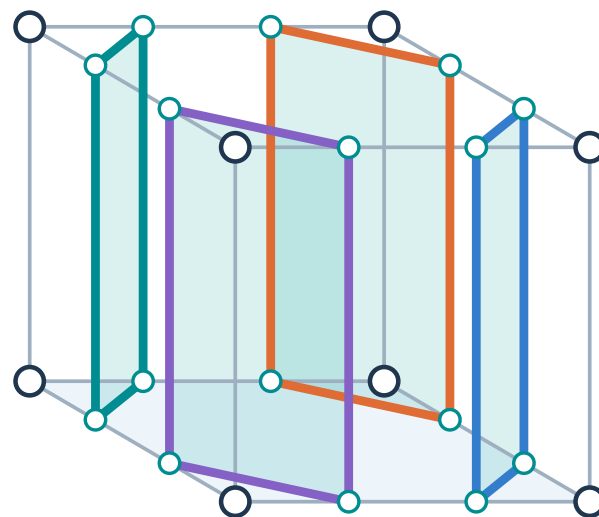


face 0x70 counts [2, 2, 2, 0]

cube 0x0AF00

CASE 1f

4 loops



face 0xF0 counts [2, 2, 2, 2]

cube 0x0FF00

面角点掩码 0000 | 双交点掩码 0111

闭环节点数: 4 / 4 / 4

连接和节点编号可在交互浏览页中对照。

面角点掩码 0000 | 双交点掩码 1111

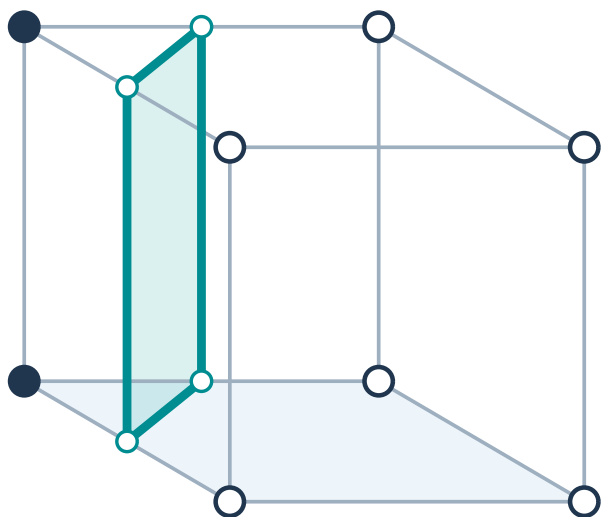
闭环节点数: 4 / 4 / 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / $2a + 2b$

CASE 2a

1 loops

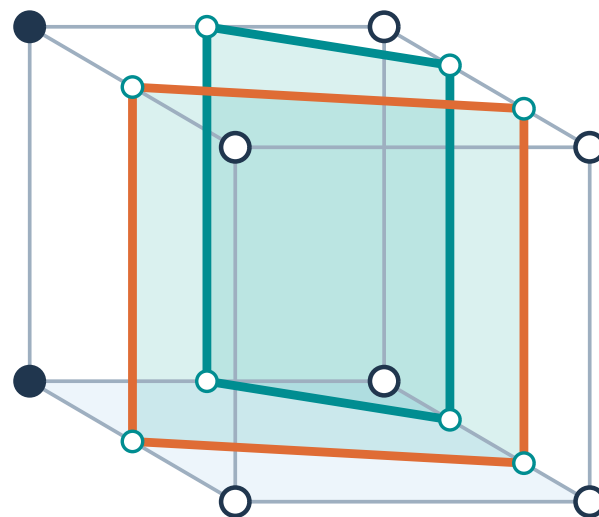


face 0x01 counts [1, 0, 0, 1]

cube 0x00011

CASE 2b

2 loops



face 0x21 counts [1, 2, 0, 1]

cube 0x0A011

面角点掩码 0001 | 双交点掩码 0000

闭环节点数: 4

连接和节点编号可在交互浏览页中对照。

面角点掩码 0001 | 双交点掩码 0010

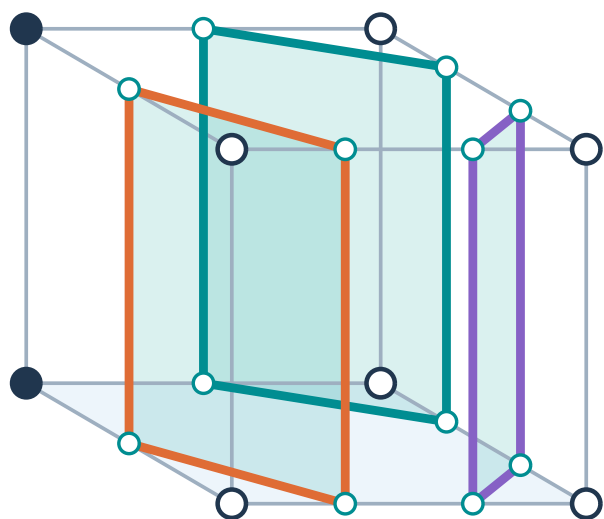
闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / $2c + 3a$

CASE 2c

3 loops



face 0x61 counts [1, 2, 2, 1]

cube 0x0AA11

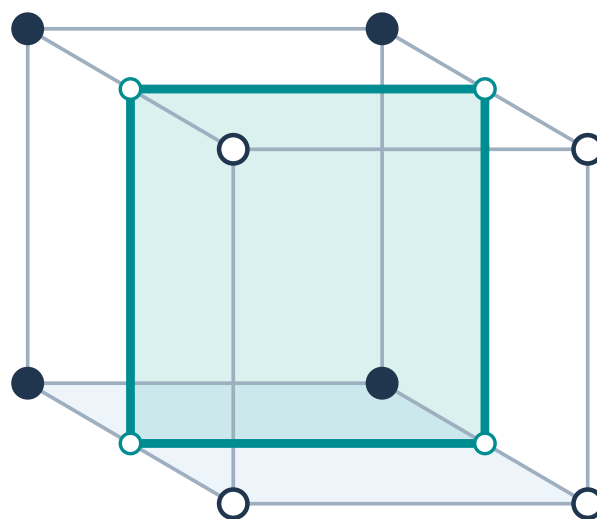
面角点掩码 0001 | 双交点掩码 0110

闭环节点数: 4 / 4 / 4

连接和节点编号可在交互浏览页中对照。

CASE 3a

1 loops



face 0x03 counts [0, 1, 0, 1]

cube 0x00033

面角点掩码 0011 | 双交点掩码 0000

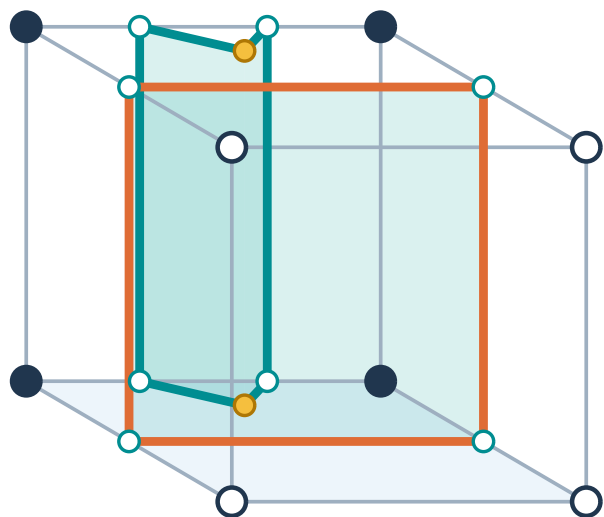
闭环节点数: 4

连接和节点编号可在交互浏览页中对照。

代表模板 / $3b + 3c$

CASE 3b

2 loops

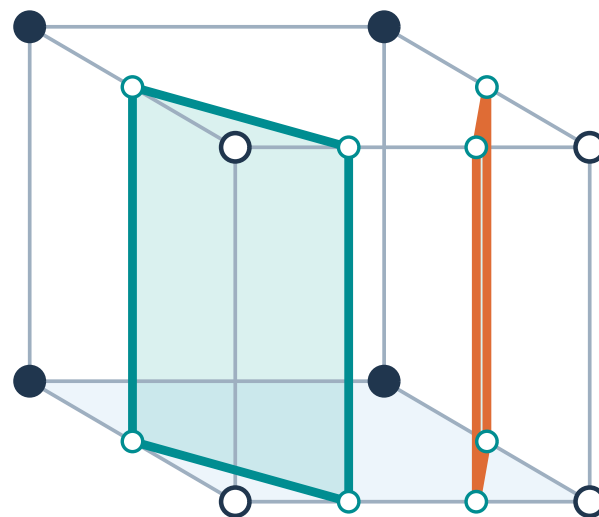


face 0x13 counts [2, 1, 0, 1]

cube 0x00533

CASE 3c

2 loops



face 0x43 counts [0, 1, 2, 1]

cube 0x00A33

面角点掩码 0011 | 双交点掩码 0001

闭环节点数: 6 / 4

黄色点为示意转折位置。

面角点掩码 0011 | 双交点掩码 0100

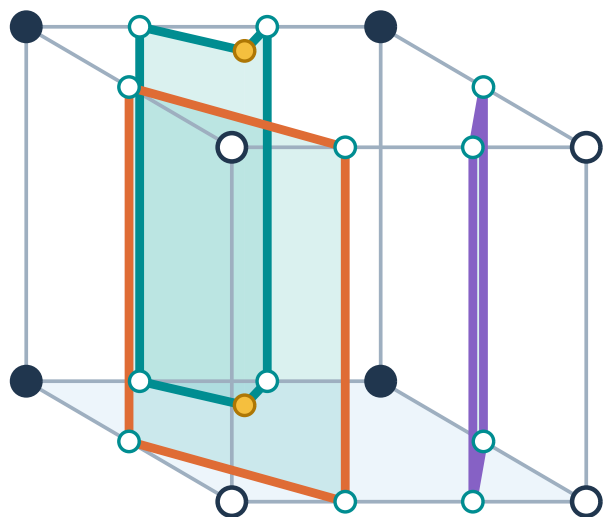
闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / 3d + 4a

CASE 3d

3 loops

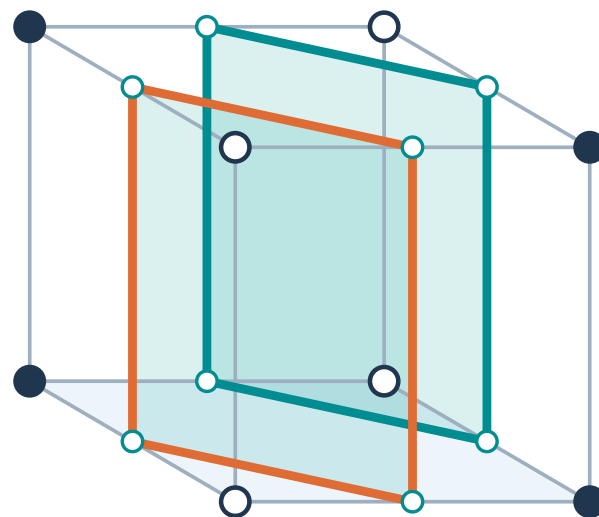


face 0x53 counts [2, 1, 2, 1]

cube 0x00F33

CASE 4a

2 loops



face 0x05 counts [1, 1, 1, 1]

cube 0x00099

面角点掩码 0011 | 双交点掩码 0101

闭环节点数: 6 / 4 / 4

黄色点为示意转折位置。

面角点掩码 0101 | 双交点掩码 0000

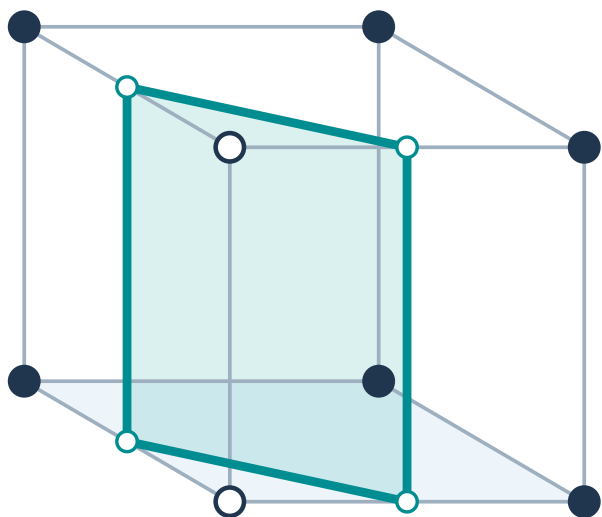
闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / 11a + 11b

CASE 11a

1 loops

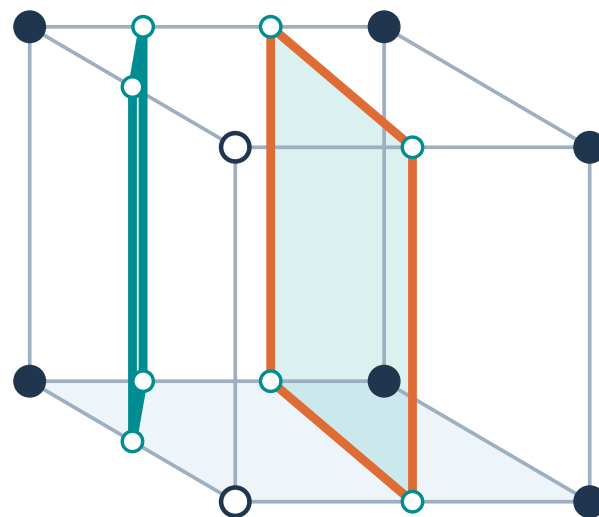


face 0x07 counts [0, 0, 1, 1]

cube 0x000BB

CASE 11b

2 loops



face 0x17 counts [2, 0, 1, 1]

cube 0x005BB

面角点掩码 0111 | 双交点掩码 0000

闭环节点数: 4

连接和节点编号可在交互浏览页中对照。

面角点掩码 0111 | 双交点掩码 0001

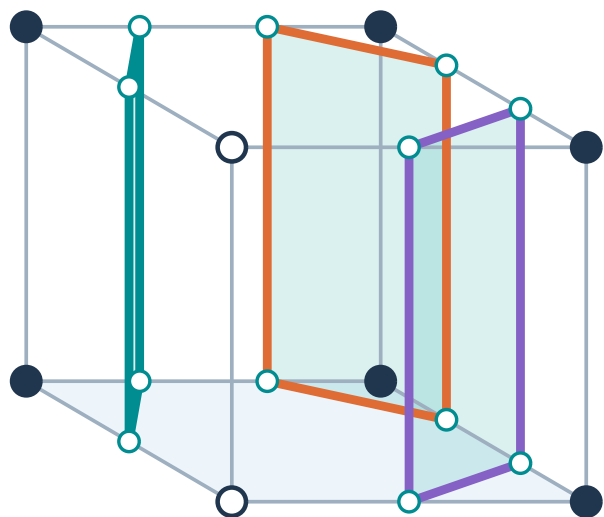
闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / 11c + 12a

CASE 11c

3 loops

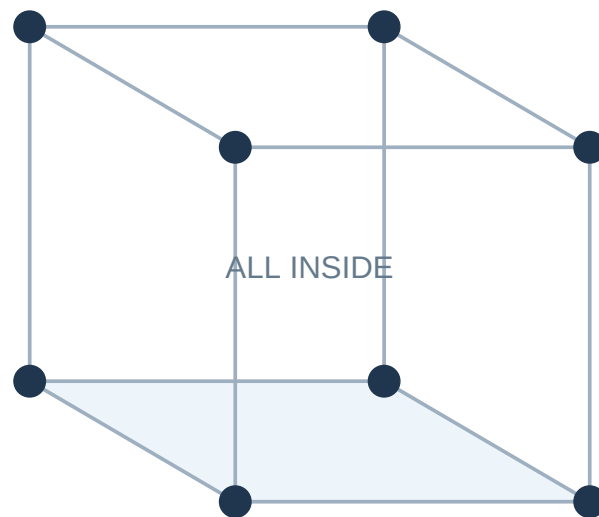


face 0x37 counts [2, 2, 1, 1]

cube 0x0A5BB

CASE 12a

0 loops



face 0x0F counts [0, 0, 0, 0]

cube 0x000FF

面角点掩码 0111 | 双交点掩码 0011

闭环节点数: 4 / 4 / 4

连接和节点编号可在交互浏览页中对照。

面角点掩码 1111 | 双交点掩码 0000

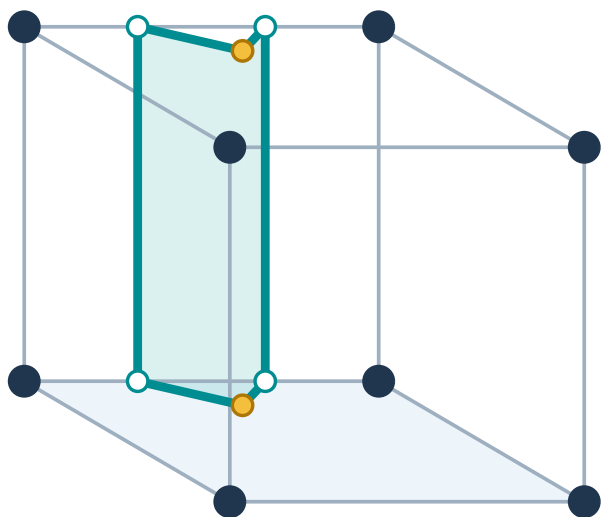
闭环节点数: 0

全部内部; 没有边界环。

代表模板 / 12b + 12c

CASE 12b

1 loops

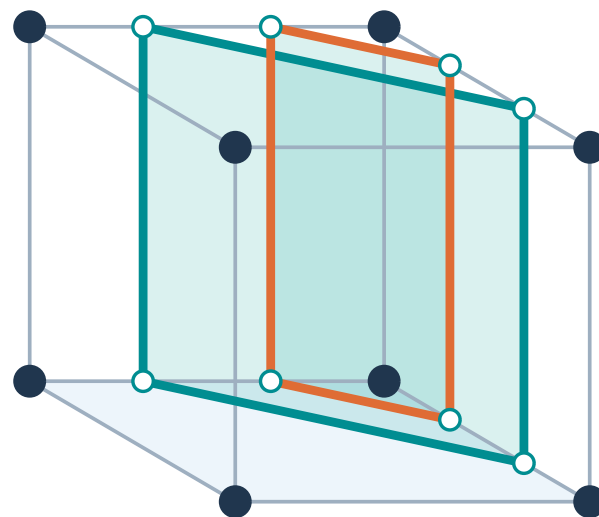


face 0x1F counts [2, 0, 0, 0]

cube 0x005FF

CASE 12c

2 loops



face 0x3F counts [2, 2, 0, 0]

cube 0x0A5FF

面角点掩码 1111 | 双交点掩码 0001

闭环节点数: 6

黄色点为示意转折位置。

面角点掩码 1111 | 双交点掩码 0011

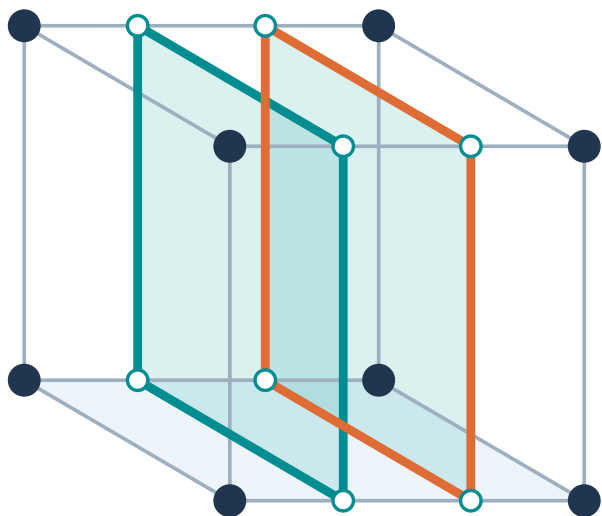
闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / 12d + 12e

CASE 12d

2 loops

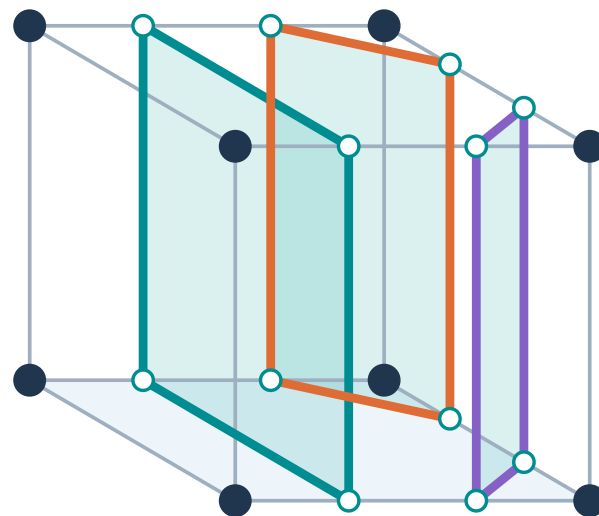


face 0x5F counts [2, 0, 2, 0]

cube 0x00FFF

CASE 12e

3 loops



face 0x7F counts [2, 2, 2, 0]

cube 0x0AFFF

面角点掩码 1111 | 双交点掩码 0101

闭环节点数: 4 / 4

连接和节点编号可在交互浏览页中对照。

面角点掩码 1111 | 双交点掩码 0111

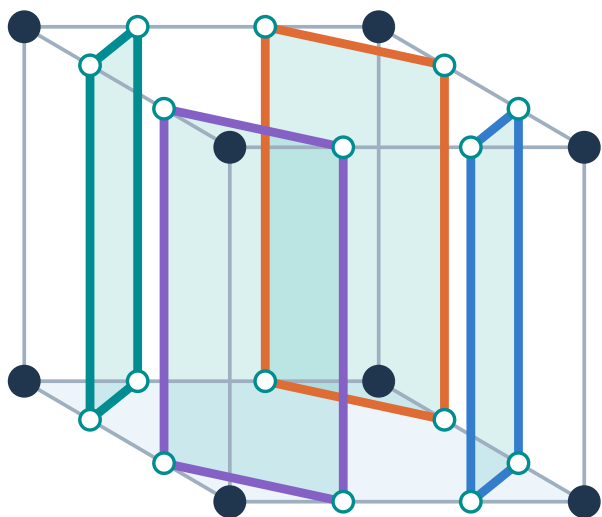
闭环节点数: 4 / 4 / 4

连接和节点编号可在交互浏览页中对照。

代表模板 / 12f

CASE 12f

4 loops



face 0xFF counts [2, 2, 2, 2]

cube 0x0FFFF

下一节：连接冲突对照

固定边交点位置，将所有角点内外状态翻转，再比较两次查表结果。全部 7 组变化会逐组列出。

面角点掩码 1111 | 双交点掩码 1111

闭环节点数：4 / 4 / 4 / 4

连接和节点编号可在交互浏览页中对照。

内外翻转时，哪些连接会改变？

这里的“冲突”指：固定面坐标与边交点位置，将四个角点的内外位全部翻转后，查表选择的无向配对改变。两侧是不同输入，不是同一个输入返回了两个矛盾答案。

全部 14 个状态均为两内两外，合为 7 对，再按旋转/镜像归成 3 类。每对闭环数量相同，仍需比较连接到哪些交点。

对照	面编码 A / B	边交点数 e0, e1, e2, e3	两侧闭环数	页码
01	0x05 / 0x0A	[1, 1, 1, 1]	2 / 2	15
02	0x13 / 0x1C	[2, 1, 0, 1]	2 / 2	16
03	0x26 / 0x29	[1, 2, 1, 0]	2 / 2	17
04	0x43 / 0x4C	[0, 1, 2, 1]	2 / 2	18
05	0x53 / 0x5C	[2, 1, 2, 1]	3 / 3	19
06	0x86 / 0x89	[1, 0, 1, 2]	2 / 2	20
07	0xA6 / 0xA9	[1, 2, 1, 2]	3 / 3	21

规则原因：当前策略在 0/1/2 个内部角点时配对外部边界区间，在 3/4 个内部角点时配对内部区间。两内两外翻转后仍有两个内部角点，所选区间可能改变。

冲突对照 01 / 0x05 与 0x0A

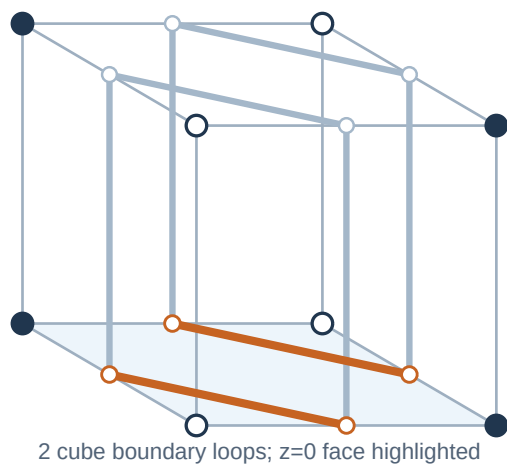
COMPLEMENT PAIR 01 / 07

family-01 / 3 D4 families

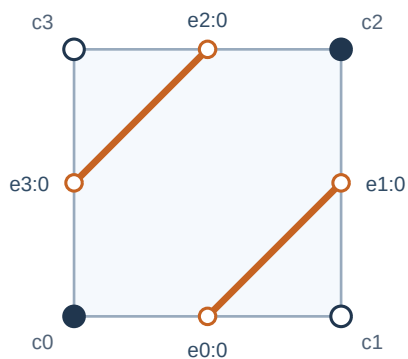
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0x05

template 4a



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

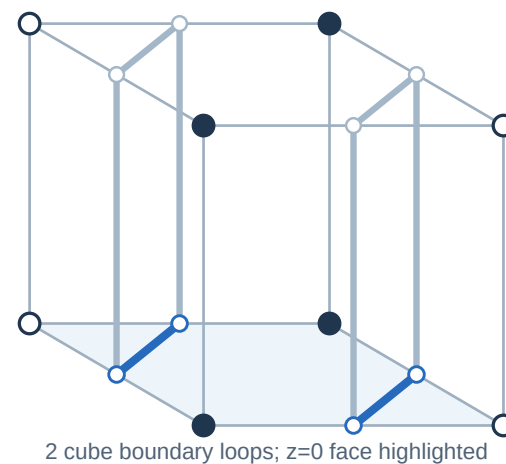
e0:0 -- e1:0

e2:0 -- e3:0

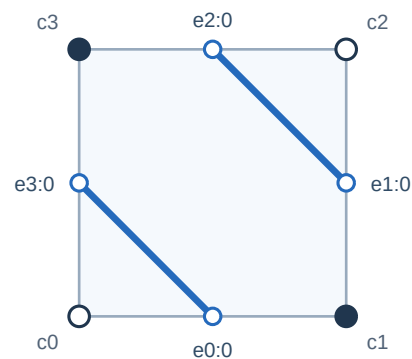
filled corner = inside
hollow corner = outside

B / face 0x0A

template 4a



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

e0:0 -- e3:0

e1:0 -- e2:0

filled corner = inside
hollow corner = outside

Diagonal corners; four single hits. Counts: [1, 1, 1, 1].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.

冲突对照 02 / 0x13 与 0x1C

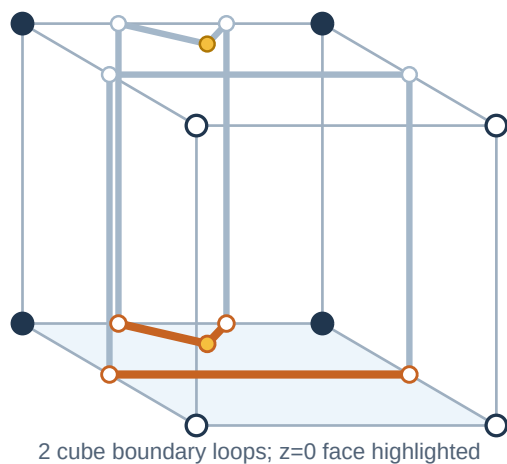
COMPLEMENT PAIR 02 / 07

family-02 / 3 D4 families

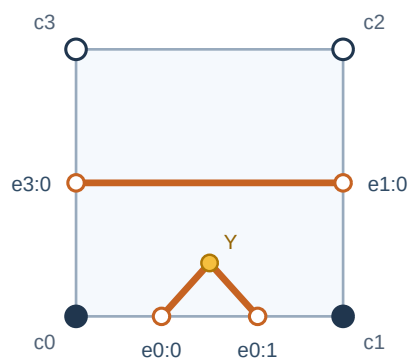
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0x13

template 3b



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



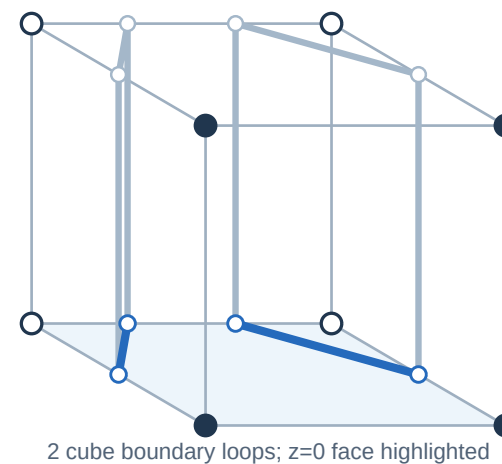
Chosen connections

e0:0 -- e0:1
via interior turn Y
e1:0 -- e3:0

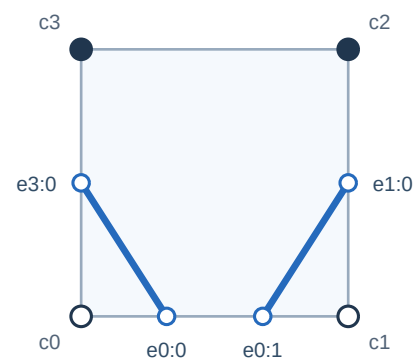
filled corner = inside
hollow corner = outside

B / face 0x1C

template 3c



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

e0:0 -- e3:0
e0:1 -- e1:0

filled corner = inside
hollow corner = outside

Adjacent corners; one double-hit edge. Counts: [2, 1, 0, 1].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.

冲突对照 03 / 0x26 与 0x29

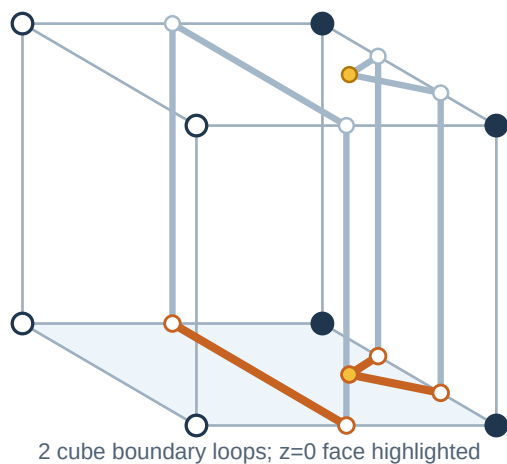
COMPLEMENT PAIR 03 / 07

family-02 / 3 D4 families

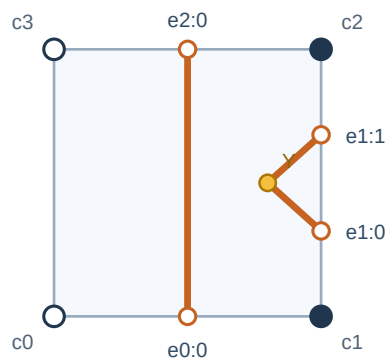
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0x26

template 3b



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



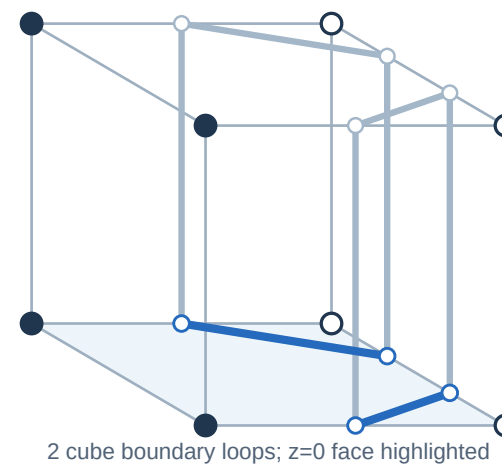
Chosen connections

e0:0 -- e2:0

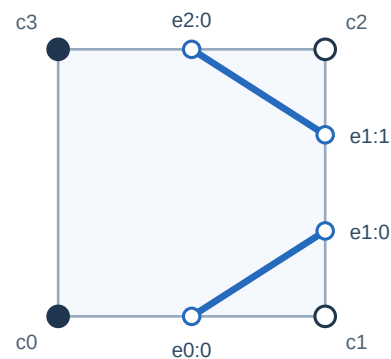
e1:0 -- e1:1
via interior turn Yfilled corner = inside
hollow corner = outside

B / face 0x29

template 3c



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

e0:0 -- e1:0

e1:1 -- e2:0

filled corner = inside
hollow corner = outside

Adjacent corners; one double-hit edge. Counts: [1, 2, 1, 0].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.

冲突对照 04 / 0x43 与 0x4C

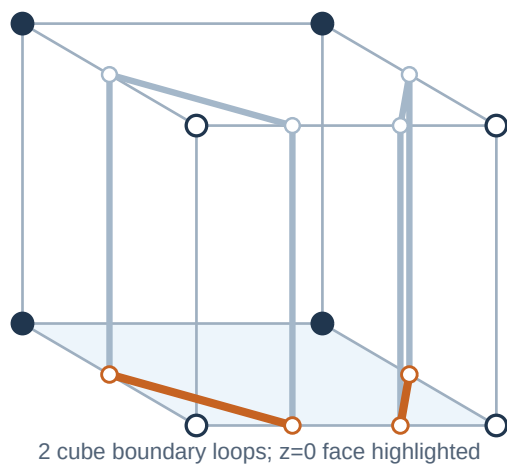
COMPLEMENT PAIR 04 / 07

family-02 / 3 D4 families

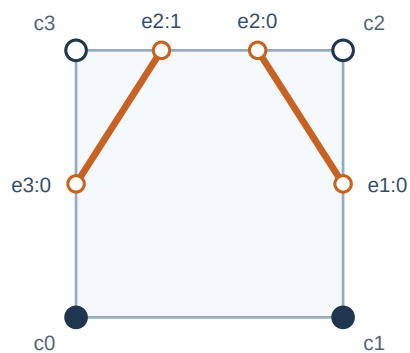
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0x43

template 3c



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

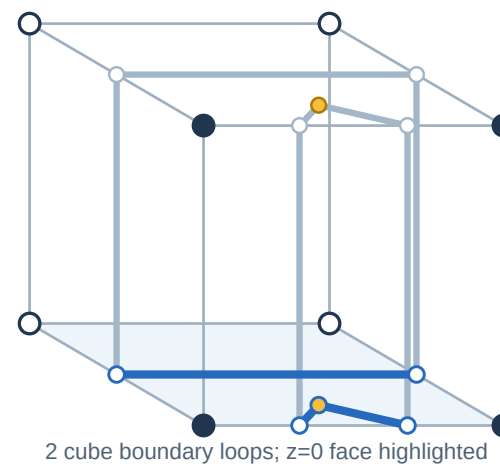
e1:0 -- e2:0

e2:1 -- e3:0

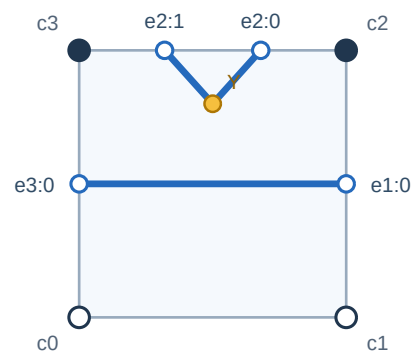
filled corner = inside
hollow corner = outside

B / face 0x4C

template 3b



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

e1:0 -- e3:0

e2:0 -- e2:1
via interior turn Yfilled corner = inside
hollow corner = outside

Adjacent corners; one double-hit edge. Counts: [0, 1, 2, 1].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.

冲突对照 05 / 0x53 与 0x5C

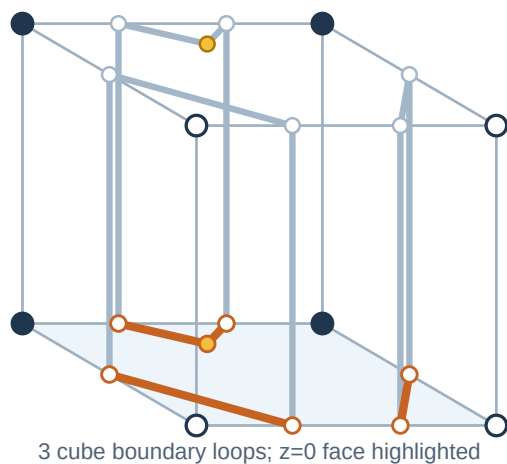
COMPLEMENT PAIR 05 / 07

family-03 / 3 D4 families

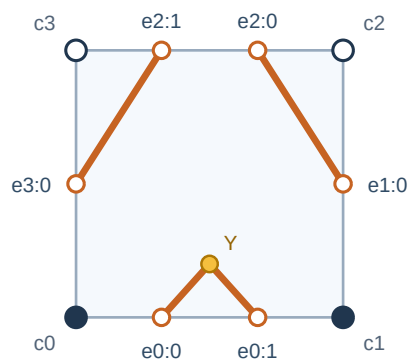
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0x53

template 3d

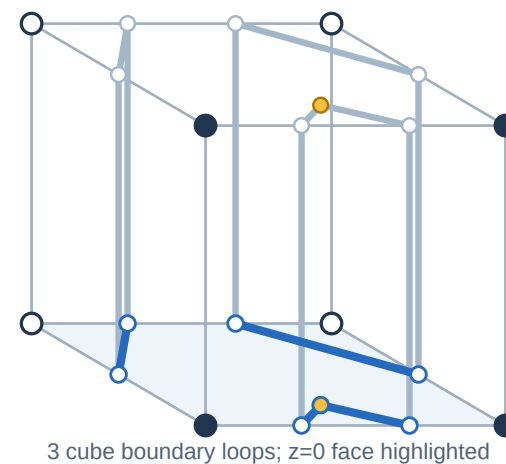


SOURCE FACE / SAME FIXED EDGE-SLOT LABELS

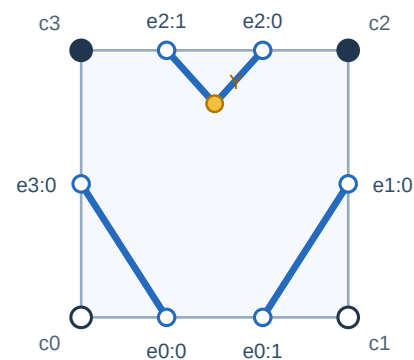


B / face 0x5C

template 3d



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Adjacent corners; two double-hit edges. Counts: [2, 1, 2, 1].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.

冲突对照 06 / 0x86 与 0x89

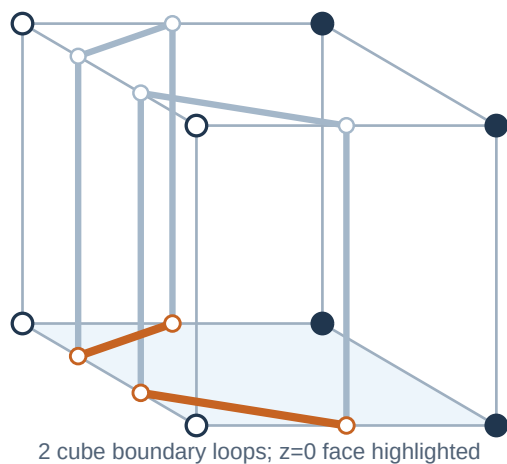
COMPLEMENT PAIR 06 / 07

family-02 / 3 D4 families

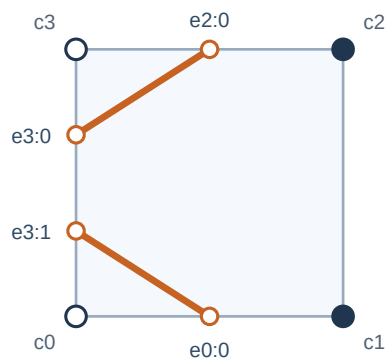
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0x86

template 3c



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

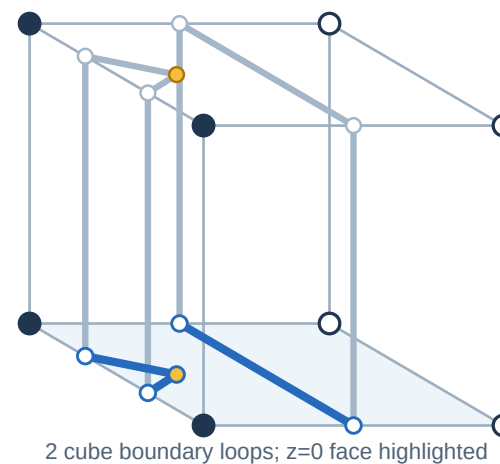
e0:0 -- e3:1

e2:0 -- e3:0

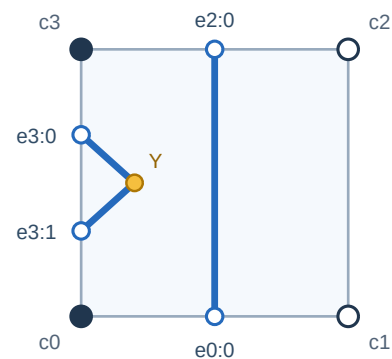
filled corner = inside
hollow corner = outside

B / face 0x89

template 3b



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

e0:0 -- e2:0

e3:0 -- e3:1
via interior turn Yfilled corner = inside
hollow corner = outside

Adjacent corners; one double-hit edge. Counts: [1, 0, 1, 2].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.

冲突对照 07 / 0xA6 与 0xA9

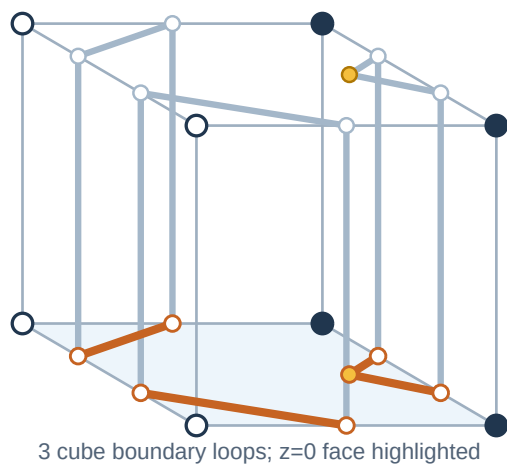
COMPLEMENT PAIR 07 / 07

family-03 / 3 D4 families

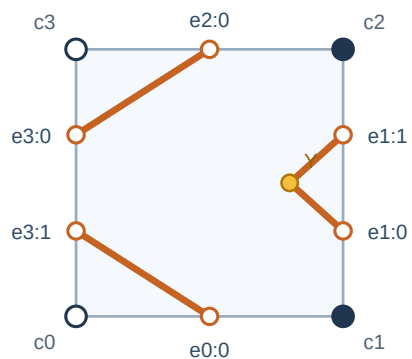
Identical edge hits; opposite corner labels; different endpoint pairing.

A / face 0xA6

template 3d



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

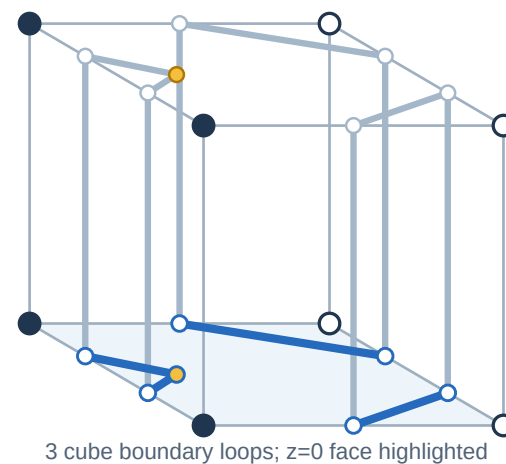
e0:0 -- e3:1

e1:0 -- e1:1
via interior turn Y

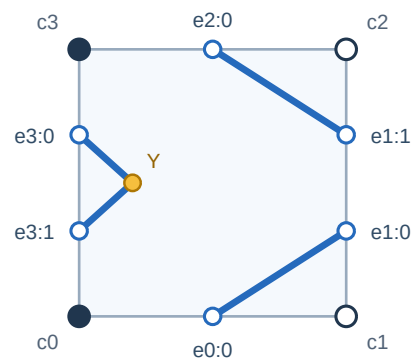
e2:0 -- e3:0

B / face 0xA9

template 3d



SOURCE FACE / SAME FIXED EDGE-SLOT LABELS



Chosen connections

e0:0 -- e1:0

e1:1 -- e2:0

e3:0 -- e3:1
via interior turn Y

Adjacent corners; two double-hit edges. Counts: [1, 2, 1, 2].

Separate complemented inputs; illustrative coordinates and turns. Boundary connectivity only, no GT or triangle reconstruction.